Information Technology and Energy Storage

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1 Introduction

This course provided me with a comprehensive overview of how energy storage and machine learning intersect, particularly in the context of modern low-power systems such as the IoT and wireless communication technologies. Throughout the course, I explored the fundamental principles behind different energy storage devices, including batteries, capacitors, and hybrid systems, and how their characteristics, such as energy density, power density, lifetime, and efficiency, impact the design and operation of embedded systems.

A key part of the course focused on understanding the energy profiles of IoT and wireless applications. I learned how factors like device duty cycles, transient power consumption, and load patterns influence the behavior of energy storage systems. Concepts such as rate-capacity effect, charge recovery, and state-of-health (SOH) management were crucial in learning why battery-powered and intermittently powered systems behave differently under varying conditions.

Another major component of the course was the application of machine learning techniques to model and optimize energy storage systems. I studied and compared several basic machine learning methods (linear regression, classification, clustering, and neural networks), and gained hands-on experience applying them to real battery datasets. Each method has unique strengths, from the interpretability of linear models to the flexibility of neural networks, and I learned how to select appropriate algorithms based on specific modeling objectives like predicting SOH, state-of-charge (SOC), or detecting battery degradation patterns.

One of the most valuable thing I gained was how artificial intelligence tools can play an important role in advancing chemical energy storage. They can be applied to improve battery lifetime prediction, optimize charging strategies, and enable smarter battery management systems.

From this course, I now have a clearer understanding of the interplay between energy storage technologies and AI-based modeling techniques, as well as how these tools can be applied to design more sustainable, efficient, and intelligent low-power systems.

2 Module M1 - Linear Regression

In this module, I learned about the fundamental principles of linear regression and how it can be effectively applied in the context of energy storage modeling. The concept of linear regression as a method used to model the relationship between a dependent variable and one or more independent variables by fitting a linear equation to the observed data. The model calculates a weighted sum of the input variables, adds a bias term, and minimizes the error between predicted and actual outcomes.

There are two main approaches to training a linear regression model. For smaller datasets, a closed-form solution can be used, which analytically computes the optimal weights by solving the normal equation. However, for larger datasets, the gradient descent method is preferred. This iterative technique updates the model parameters by computing the gradients of a loss function—commonly the mean squared error—and adjusting the weights and bias until the model converges to the optimal solution. Regularization techniques such as Lasso (L1 regularization) and Ridge regression (L2 regularization) can also be used to prevent overfitting by penalizing large coefficients.

This module also emphasized the importance of feature engineering techniques, such as feature selection, feature extraction, and feature transformation, which are essential to improve model performance and interpretability.

2.1 Cross-temperature Batteries SOC(State-of-Charge) Prediction

In this study, we focus on cross-temperature prediction of lithium-ion batteries' State-of-Charge (SOC) using a linear regression model. Lithium-ion batteries are crucial in electric vehicles and the energy sector, where accurate SOC estimation ensures safety and efficiency. The SOC dataset used in our analysis is derived from the DST (Dynamic Stress Test) measurements, covering various temperatures (0°C, 25°C, 50°C), with features such as current, voltage, temperature, and SOC. Specifically, we utilized data from the MambaLithium study [1], which is publicly available on GitHub (Link). This dataset has been consistently used across all modules throughout the course for experimental purposes.

We trained a linear regression model using the 25°C data and evaluated its performance when applied to lower (0°C) and higher (50°C) temperatures. Our key findings include:

(1) Model Generalization: The linear model trained at 25°C generalizes well to both 0°C and 50°C, particularly performing better at 50°C. This may be due to the smoother battery discharge behavior at extreme temperatures, leading to a more linear SOC trend, which the model captures effectively.

(2) End-of-Discharge Challenge: A issue arises near the battery's deep discharge phase, where SOC drops sharply, particularly at low temperatures. The linear model struggles to accurately predict this behavior, as shown in the figure below.

(3) Linear vs. Non-linear Relationship: Despite including temperature as a feature, the model cannot fully capture the non-linear SOC dynamics during late-stage discharge, this is a limitation of model complexity of linear regression.

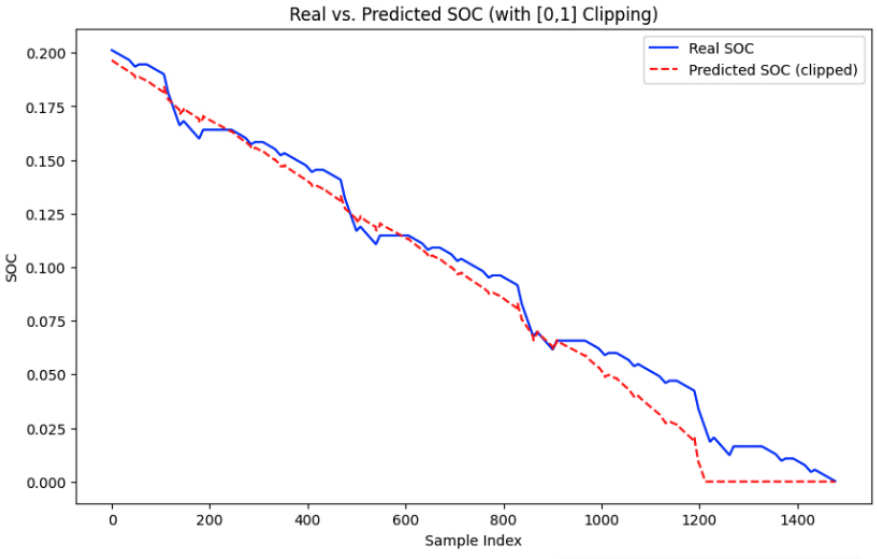


Figure 1: Real vs. Predicted State of Charge (SOC) with linear regression

2.2 Key Insights

In our study of SOC estimation using both linear regression and LSTM model (in later modules), we observed that towards the end of the dataset, the predicted SOC occasionally dropped sharply—sometimes even nearing zero—despite the actual SOC remaining relatively stable. This discrepancy indicates that the model tends to equate transient drops in voltage with significant depletion of the battery, which result in an downward trend in the predicted curve. See Figure 2.

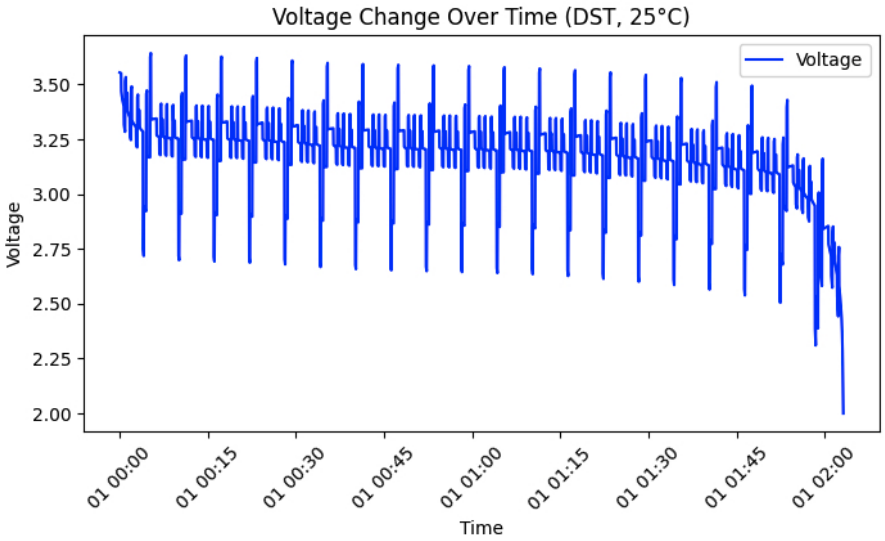


Figure 2: In the raw data, the voltage drops with time

One primary reason for this behavior we think may be from the models' overly simplified assumption that a drop in voltage always corresponds to an equivalent drop in SOC. In reality, the battery's terminal voltage can fluctuate for various reasons—such as temperature effects, polarization, and rest intervals—without reflecting a truly proportional loss in remaining capacity. Without additional features to distinguish transient voltage dips from genuine depletion, the models automatically interpret any voltage reduction as a direct indicator of substantial SOC decline.

When we learned about other group's study during the seminar, I noticed that some groups applied linear regression models without conducting sufficient preprocessing on their input features. It seemed that certain irrelevant or suboptimal features were included without careful selection, which could potentially limit the model's predictive performance. Also, features with differing scales (such as current, voltage, and temperature) were used directly, without normalization or standardization. This might affect the model's ability to assign appropriate weights to each feature, especially when features with larger numerical ranges dominate the learning process. Therefore, implementing feature selection techniques and applying normalization could help improve the robustness and generalization of linear regression models in SOC estimation tasks.

2.3 Modeling Energy Storage

Linear regression can be applied to model real-world energy storage systems. For example, in the context of battery performance prediction, linear regression can be used to model the degradation of battery capacity over time. Variables such as the number of charge-discharge cycles, temperature, charging rate, and depth of discharge can serve as independent variables. The remaining capacity of the battery can be treated as the dependent variable. By training a linear regression model on historical operational data, it becomes possible to predict future battery degradation trends. This not only aids in optimizing battery usage and maintenance schedules but also extends the overall lifespan of energy storage systems. What I found particularly valuable is that linear regression provides an interpretable and computationally efficient framework, making it well-suited for practical applications in chemical energy storage modeling.

3 Module M2 - Classification

In this module, I learned about the fundamental principles of classification within supervised machine learning. The goal of classification is to assign input data points to specific predefined categories based on their features, allowing us to make meaningful predictions about unseen data. There are two main approaches to achieving this. Instance-based classification methods, like K-NN, make predictions by calculating the distance between a query point and existing labeled instances, selecting the majority class among the nearest neighbors. This approach is straightforward and effective for smaller datasets but can be computationally intensive for larger ones.

On the other hand, model-based methods build an explicit model during the training phase. Decision trees, for example, recursively split the data based on feature conditions, using metrics like information gain or Gini impurity to determine the best splits. Random forests improve upon decision trees by creating an ensemble of trees trained on different subsets of data and features, reducing variance and overfitting while increasing model robustness. Essential evaluation metrics, including accuracy, precision, recall, and F1-score, are used to assess the performance of classification models.

3.1 SOC Classification With Two Models

In this study, we focus on SOC classification. We categorized the battery into "low," "medium," or "high" states by defining a simple function `soc_to_class`, which assigns a numeric class label (0, 1, or 2) based on the SOC thresholds (e.g., $\texttt{SOC} < 0.3$, $0.3 \leq \texttt{SOC} < 0.7$, $\texttt{SOC} \geq 0.7$). We used K-Neighbors Classifier because k-NN relies on distance metrics, scaling features with MinMaxScaler ensures that "Current," "Voltage," and "Temperature" are on comparable scales. We also created a Decision Tree Classifier with a max depth of 5 just as an example. We computed accuracy, a confusion matrix, and a classification report (precision, recall, and F1-score). These metrics help see if the classifier is favoring one SOC range over another or if it's balanced. See Figure 3.

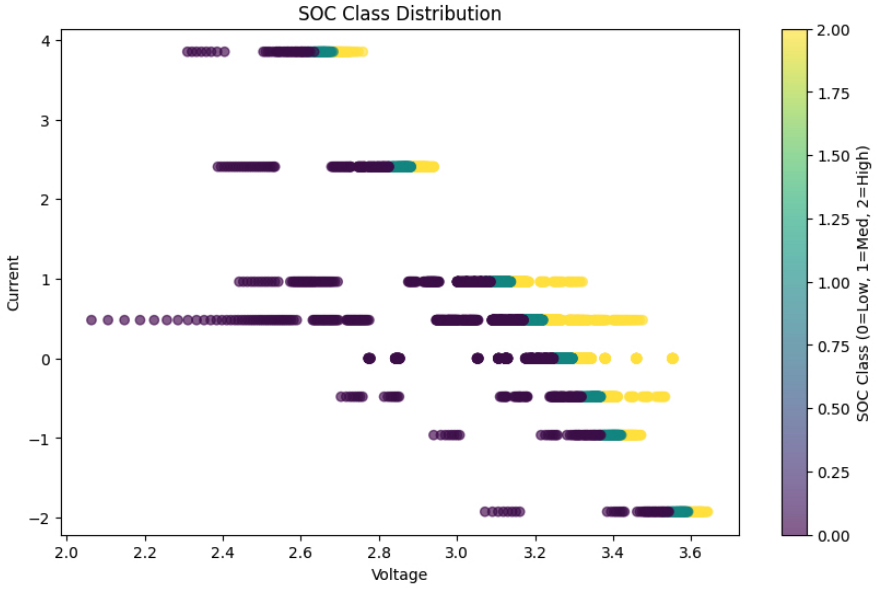


Figure 3: This plot shows the battery's State of Charge (SOC) categorized into three classes (Low, Medium, and High) as a function of voltage on the x-axis and current on the y-axis, recorded under 25°C conditions.

With an accuracy over 90% for both models, it's clear that grouping SOC into Low/Medium/High is tractable using these simple algorithms. Depending on application needs, k-NN provides a higher overall score in this dataset, while the tree-based method can be advantageous if explainability or decision rules are desired.

3.2 Key Insights

In our study, we observed that K-NN outperforms the Decision Tree in terms of overall accuracy (95% vs. 91%) and shows higher precision/recall across the classes. The Decision Tree still performs respectably, but struggles more in differentiating certain classes—particularly class 2 (High SOC). The data appear well-suited to distance-based classification (k-NN), which may indicate that "Current," "Voltage," and "Temperature" form relatively distinct clusters for each SOC range. The Decision Tree might need more tuning (e.g., max_depth, min_samples_split) or additional features to reduce confusion between classes.

From the classification results, each class shows its own precision, recall, and F1:

Table 1: Classification Result (k-NN)

Class	Precision	Recall	F1-score	Support
0	0.97	0.96	0.97	449
1	0.92	0.96	0.94	560
2	0.97	0.94	0.96	469
Accuracy			0.95	1478
Macro Avg	0.96	0.95	0.95	1478
Weighted Avg	0.95	0.95	0.95	1478

Table 2: Classification Result (Decision Tree)

Class	Precision	Recall	F1-score	Support
0	0.95	0.93	0.94	449
1	0.85	0.96	0.90	560
2	0.97	0.84	0.90	469
Accuracy			0.91	1478
Macro Avg	0.92	0.91	0.91	1478
Weighted Avg	0.92	0.91	0.91	1478

(1) for K-NN: Class 0 has near 97% precision and 96% recall. Class 1 and class 2 both also exceed 90% for precision and recall. Weighted-average F1 is 0.95.

(2) for Decision Tree: Class 0 precision is 0.95, recall 0.93; Class 1 recall is quite high (0.96) but precision is lower (0.85); Class 2 suffers more confusion, recall around 0.84; Weighted-average F1 is 0.91.

Recall indicates how many samples of a given class were correctly identified, while precision measures how many predictions for that class were correct. The F1-score balances both. Here, k-NN is more balanced across classes; the Decision Tree tends to have stronger performance for classes 0 and 2 on certain metrics, but has lower recall for class 2 overall.

When we learned about other group's study during the seminar, I noticed some imbalanced class distributions. Real-world battery datasets may concentrate heavily around certain SOC or SOH ranges (e.g., mostly mid-to-high SOC during typical usage), this will lead to to uneven classes. Classifiers can become biased if there are too few samples in a "rare" category, potentially overlooking critical failure or aging events.

3.3 Modeling Energy Storage

The application of classification I explored in modeling energy storage is battery health monitoring. Specifically, classification models can be used to predict whether a battery is in "Good" condition or requires replacement, based on features such as capacity, temperature, charge-discharge cycles, and charging behavior. For instance, decision tree models can effectively capture decision rules from historical battery performance data, providing interpretable insights into which factors most influence battery degradation. Random forests further enhance this application by delivering more accurate and stable predictions through ensemble learning. What I found particularly compelling is that using classification models allows for proactive maintenance and safer operation of energy storage systems, ensuring that faulty batteries are identified and replaced before causing performance issues or safety risks.

4 Module M3 - Clustering

Clustering is an unsupervised machine learning technique used to group similar data points based on shared characteristics. The goal of clustering is to discover inherent structures or groupings within a dataset without the need for labeled examples. In doing so, clustering helps to uncover hidden patterns, reduce data dimensionality, detect anomalies, and enhance interpretability.

The main approaches include prototype-based methods like K-Means, which rely on centroids to represent clusters; density-based methods such as DBSCAN, which focus on areas of high point density; and hierarchical clustering, which creates nested clusters organized in a tree structure. Each method has its strengths and limitations. For example, K-Means is simple and efficient but struggles with irregularly shaped clusters and requires prior knowledge of the number of clusters. DBSCAN addresses some of these limitations by allowing for non-globular clusters and detecting noise points. Techniques such as the elbow method and silhouette score were discussed to help evaluate clustering quality and determine the optimal number of clusters.

4.1 Clustering Analysis of Battery Data Using K-Means and DBSCAN

In this module, we explored unsupervised learning techniques, specifically focusing on clustering methods and dimensionality reduction. we began by examining two widely used clustering algorithms: K-Means and DBSCAN. Through the application of K-Means, we learned how to partition data into distinct groups by minimizing within-cluster variance, and how to utilize tools such as the elbow method to determine the optimal number of clusters. Additionally, DBSCAN provided me with a contrasting perspective, as it identifies clusters based on density rather than predefined numbers, making it particularly effective for detecting noise and handling data with varying cluster shapes. By comparing these two approaches, we gained a clearer understanding of their respective strengths: K-Means excels in cases where clusters are well-separated and spherical, while DBSCAN is more robust in dealing with complex, non-globular structures.

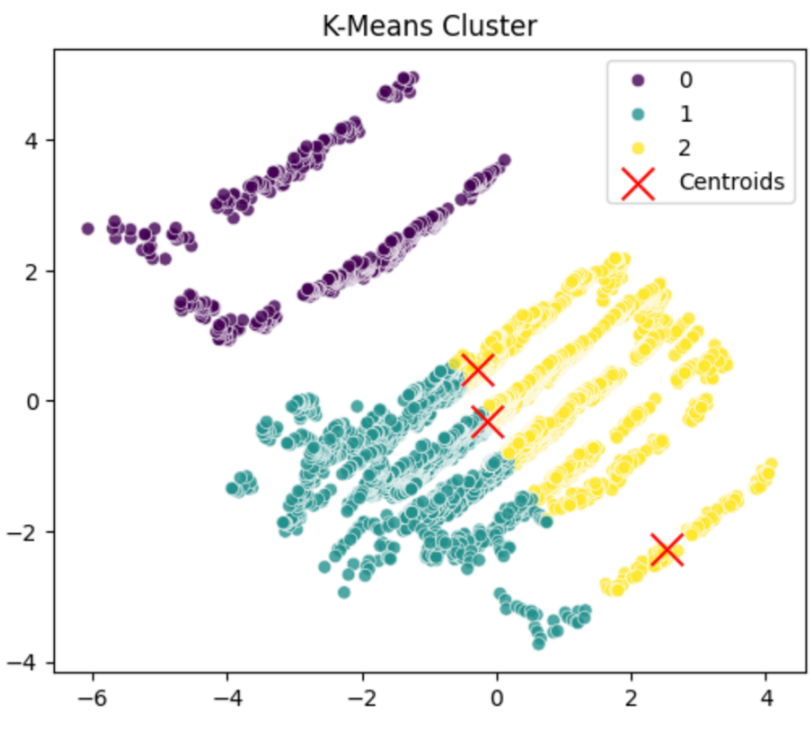


Figure 4: K-Means Cluster groups battery data into 3 clusters

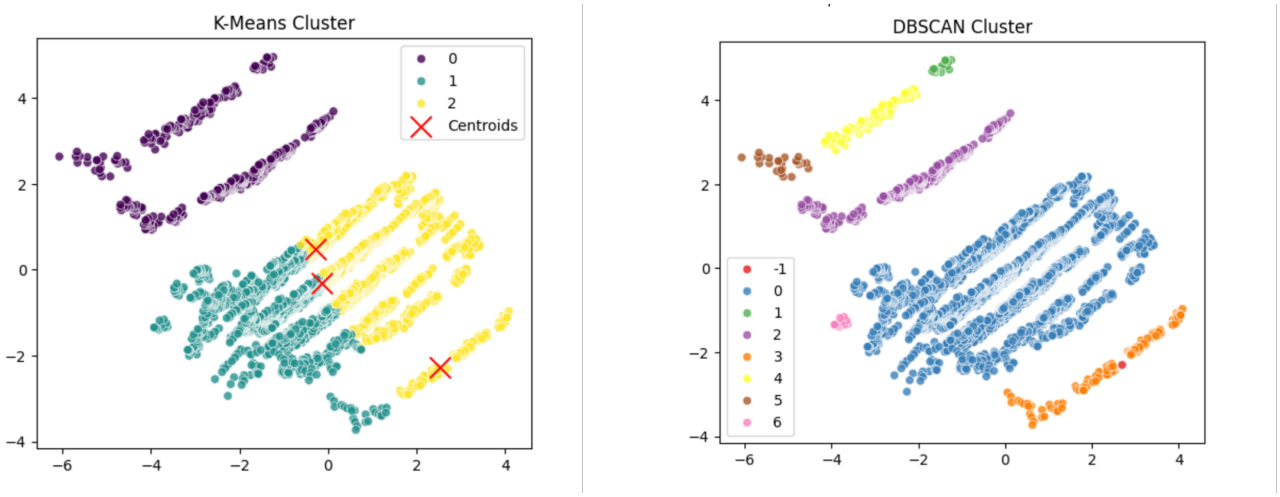
4.2 Key Insights

Beyond clustering, we explored dimensionality reduction techniques to simplify and better visualize high-dimensional data. Principal Component Analysis (PCA) was a key method we applied. Additionally, we used feature selection techniques, such as analyzing feature importance and variance, to identify and retain the most relevant features.

K-Means provides a straightforward way to partition the data into a specified number of clusters, and its results indicate moderately distinct groupings, as suggested by the silhouette score. However, K-Means relies on distance

from cluster centroids, making it less flexible when clusters have irregular shapes or different densities.

In contrast, DBSCAN automatically discovers arbitrarily shaped clusters and flags outliers as noise points. This adaptability is particularly advantageous for data where clusters vary in size and density, although DBSCAN's performance hinges heavily on choosing an appropriate value for epsilon (eps).



(a) K-Means Cluster

(b) DBSCAN Cluster

Figure 5: Comparison of K-Means and DBSCAN Clustering

From the boxplot of SOC across the three identified clusters, we observed that Cluster 0 exhibits a broad range of SOC values, suggesting a battery group with stable performance and wide operational coverage. Cluster 1 tends to have consistently lower SOC, implying an aging battery or lower efficiency state. Meanwhile, Cluster 2 maintains generally higher SOC levels, indicating a potentially more efficient or better-maintained battery set. These observations highlight how clustering can reveal distinct operational profiles within the dataset, which may guide targeted maintenance or usage strategies.

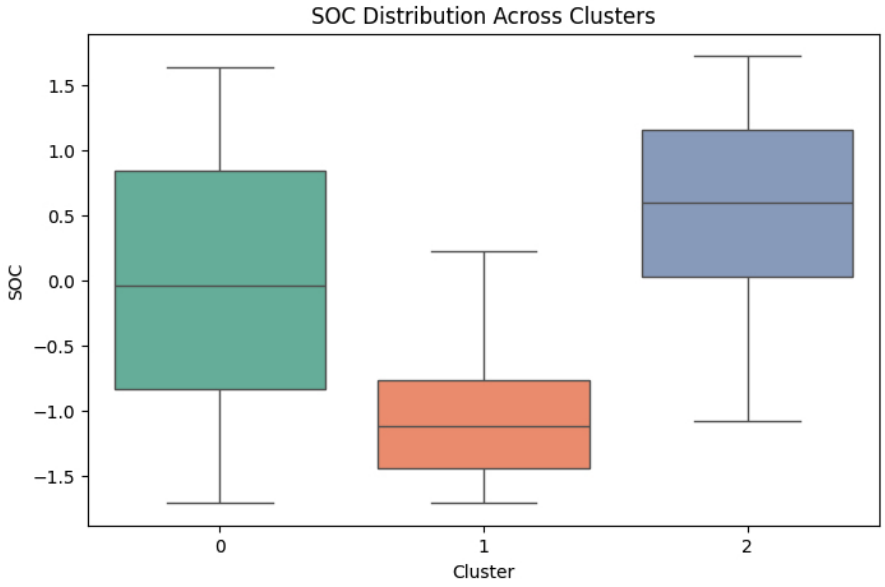


Figure 6: SOC Distribution Across Clusters

When we learned about other group's study during the seminar, I observed that battery datasets often contain multiple types of features (current, voltage, temperature, cycles, etc.) and may span a wide range of conditions. High-dimensional or highly diverse data can dilute distance metrics, making it harder for clustering algorithms to discover meaningful groupings. Sensor noise, measurement inaccuracies, and data outliers can complicate clustering. Even density-based methods can struggle if noise levels are high enough to blur boundaries between clusters.

4.3 Modeling Energy Storage

One practical application of clustering that I learned in the context of modeling energy storage is the identification of different battery health states. For instance, battery cells can exhibit varying behaviors based on factors such as voltage, capacity, and state of health (SOH). By applying clustering techniques, we can group battery cells with similar performance characteristics into clusters representing different health conditions—such as "Healthy," "Moderate," or "Degraded." This unsupervised approach is particularly valuable when labeled data on battery conditions is unavailable.

For example, using K-Means clustering allows us to segment large datasets of battery operational data into meaningful categories, which can then guide maintenance strategies or predictive monitoring. Moreover, density-based methods like DBSCAN can help identify outlier battery cells that may indicate early signs of failure, contributing to safer and more efficient energy storage system management. What I appreciated most is how clustering enables data-driven decision-making without requiring prior labeling, making it highly applicable to real-world chemical energy storage problems where large volumes of unlabeled operational data are common.

5 Module M4 - Neural Networks

The primary goal of artificial neural networks is to model complex, nonlinear relationships between inputs and outputs, allowing for highly flexible and accurate predictions even when traditional linear models fail. ANNs achieve this by applying nonlinear activation functions to transform inputs and learning hierarchical feature representations through multiple layers. The basic approach starts with the Multi-Layer Perceptron (MLP), where data passes through several hidden layers, and the model parameters are optimized via backpropagation and gradient descent. Activation functions such as ReLU, Sigmoid, or Tanh play a crucial role in introducing nonlinearity, while techniques like binary cross-entropy loss and weight updates ensure the network learns effectively. Furthermore, more complex approaches like CNNs and LSTMs extend the capabilities of standard ANNs by incorporating spatial and temporal data processing, making them suitable for a wider range of applications.

5.1 SOC Prediction Using CNN and LSTM

In our study, we explored two different neural network architectures (CNN and LSTM) to estimate the State of Charge (SOC) for a battery dataset measured under 25°C conditions. The CNN approach involved convolving over short time windows (10 time steps) to capture local temporal patterns, followed by pooling and fully connected layers. After tuning hyperparameters like learning rate, dropout, and regularization, the CNN achieved smoother validation loss and reasonably accurate SOC predictions, especially in the high and low SOC ranges. Meanwhile, the LSTM method, which I implemented with a window of 10 consecutive time steps, demonstrated the capacity to track SOC trajectories over time and produced predictions that aligned relatively well with the ground truth, though the model tended to underestimate SOC at the deeper discharge range. See Figure 7.

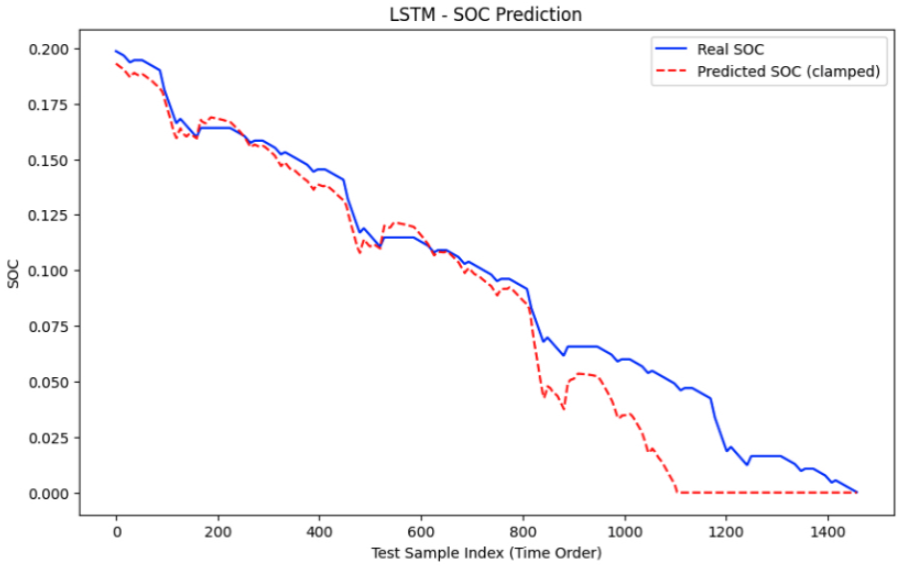


Figure 7: Real vs. Predicted State of Charge (SOC) with LSTM

5.2 Key Insights

One insight was that how sensitive both architectures were to subtle voltage drops in the latter portion of the discharge. Small shifts in voltage sometimes led to disproportionately large changes in predicted SOC, suggesting that the models were over-relying on instantaneous voltage cues without fully incorporating longer-term battery state evolution.

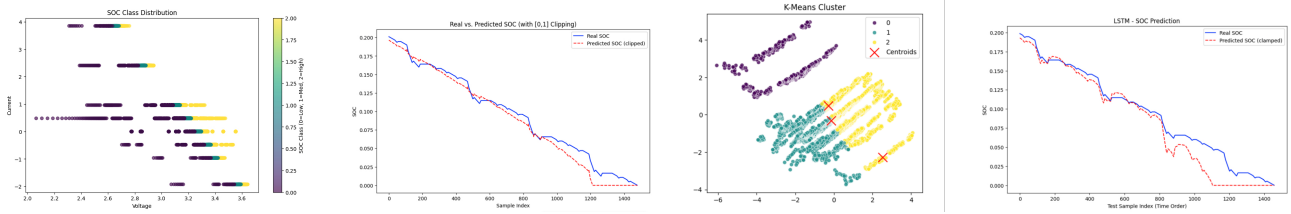
Although CNN is well-suited to learning local time-window patterns, it occasionally struggled to place short-term voltage fluctuations in the broader context. The LSTM, with its inherent gating mechanisms and memory cells, proved slightly more robust in smoothing out transient dips in voltage. However, neither method entirely overcame the tendency to latch on to subtle voltage shifts, demonstrating that deeper sequence modeling (perhaps via multi-layer LSTMs or incorporating domain-specific features) could further improve performance.

A potential limitation I noticed from other groups was: with both neural network approaches is their reliance on representative training data across the entire SOC range, particularly at extreme values. If the dataset had fewer samples where SOC was close to zero or near 100%, the models tended to extrapolate poorly in those regions. This challenge underscores not only the need for richer data but also for architectures that can adapt or incorporate

physical battery models to prevent large prediction errors when the network encounters boundary conditions or underrepresented states.

5.3 Modeling Energy Storage

One application of neural networks I explored in the context of modeling energy storage is battery performance prediction. Battery systems exhibit highly nonlinear behavior influenced by various factors such as temperature, charge-discharge cycles, and material properties. ANNs can capture these complex interactions to predict battery lifetime, efficiency, or capacity degradation without the need for exhaustive experimental testing. For example, LSTM networks are particularly well-suited for analyzing time-series data from battery sensors, enabling accurate estimation of state-of-charge (SOC), forecasting of degradation trends, and prediction of remaining useful life (RUL). CNNs can be applied when dealing with spatial data, such as thermal imaging of battery cells, to detect potential defects or anomalies. What I found especially valuable is that ANNs significantly accelerate battery research and development by reducing the dependency on time-consuming experiments, allowing for real-time optimization and smarter battery management systems in applications like electric vehicles and large-scale energy storage systems.



(a) Classification: SOC categorized into three classes as a function of voltage

(b) Linear Regression: Real vs. Predicted SOC with linear regression

(c) Clustering: K-Means groups battery data into 3 clusters

(d) Neural Network: Real vs. Predicted SOC with LSTM model

Figure 8: Comparison of the four machine learning methods covered in the course.

6 Discussion and Conclusion

6.1 Comparison of Machine Learning Methods

In this course, I learned about four fundamental machine learning methods (linear regression, classification, clustering, and artificial neural networks). Linear regression is a supervised learning method aimed at modeling the linear relationship between dependent and independent variables. Its goal is to predict continuous outcomes by fitting a weighted sum of input features to minimize prediction error. Classification, another supervised method, focuses on assigning discrete labels to input data based on learned decision boundaries. It can be implemented using instance-based approaches or model-based approaches. Clustering, an unsupervised learning technique, aims to group similar data points into clusters based on shared features without requiring labeled data. Lastly, artificial neural networks are powerful models capable of learning complex, nonlinear relationships through interconnected layers of neurons, optimized via backpropagation.

Each of the four machine learning methods discussed in the course contributes uniquely to different questions of battery and energy storage systems. (i) For transient response, linear regression and ANNs are quite effective. Linear regression can provide interpretable models of how input variables like current or temperature influence voltage or power response over short timescales. ANNs, with their ability to capture nonlinear dynamics, are especially well-suited for accurately modeling transient behaviors in systems with complex electrochemical processes. (ii) Regarding state of charge (SOC) estimation, classification models such as decision trees and K-NN can classify SOC levels into discrete categories (e.g., low, medium, high), while ANNs, particularly LSTM networks, can represent time-series sensor data for SOC predictions. (iii) For state of health (SOH) monitoring, clustering methods are highly valuable. By analyzing patterns in historical operational data, clustering algorithms can segment batteries into groups representing different health conditions or detect anomalous behavior indicative of degradation. Classification models can also be used to categorize batteries as "healthy" or "needs replacement" based on key features. (iv) In the analysis of material composition, clustering and ANNs offer complementary strengths. Clustering can help in grouping materials with similar electrochemical properties, aiding in material discovery and optimization, while ANNs can predict material properties based on compositional data, accelerating the screening of new battery materials.

Each machine learning method has its strengths and limitations. Linear regression is highly interpretable and computationally efficient, making it suitable for scenarios where linear relationships dominate, such as modeling basic transient responses. However, its inability to capture complex nonlinearities limits its applicability in more intricate systems. Classification methods offer simplicity and ease of implementation, making them effective for discrete decision-making tasks like SOC classification or basic SOH labeling. However, their reliance on predefined categories can reduce flexibility when dealing with continuous or highly dynamic data. Clustering methods is suitable for unlabeled data, such as detecting degradation patterns or grouping batteries with similar characteristics, but they may struggle with high-dimensional, noisy data and require careful parameter selection. ANNs are powerful and flexible, capable of complex and nonlinear relationships, making them ideal for tasks like real-time SOC prediction, modeling transient response, and analyzing material compositions. However, ANNs require large amounts of training data, substantial computational resources, and careful tuning to avoid overfitting.

A synergistic approach combining these methods can offer significant benefits. For example, clustering can be used as a preprocessing step to segment battery datasets, followed by ANN models trained separately on each cluster to improve accuracy and reduce computational load. Classification models can be layered on top of clustering results to provide decision-making logic for battery health management. Linear regression can be integrated within larger ANN frameworks for the initial analysis of the importance of characteristics. I recommend using linear regression and classification methods for interpretable, low-complexity tasks such as basic SOC or SOH assessments, while utilizing ANNs and clustering techniques for more advanced tasks like real-time performance

prediction and degradation forecasting.

6.2 Energy Storage in the Context of Low-Power Communicaiton and Compuation

Through this course, I gained a deeper understanding of energy storage devices and their practical implications, particularly in the context of low-power communication and computation. I learned about the operation and characteristics of modern energy storage systems, including batteries, capacitors, and hybrid devices. I explored macroscopic phenomena such as the rate-capacity effect and charge recovery, both of them are important when managing energy storage for embedded systems and IoT applications.

One key understanding from the course is how energy storage limitations affect the design of low-power communication and computing systems. Traditional battery-powered devices face significant challenges: batteries are bulky, degrade over time, and require maintenance. They are not suitable for applications that require long-term sustainability or deployment in remote areas. This led me to understand intermittently powered systems. In such systems, energy harvesting from ambient sources like RF, vibration, or solar replaces the need for batteries. However, intermittent availability of energy faces unique challenges. Devices must handle unpredictable power failures, maintain system state through checkpointing mechanisms, and ensure forward progress of computation despite interruptions. Systems like Mementos and Mayfly, which introduce runtime support and task-based programming models, are innovative solutions that allow devices to operate efficiently under intermittent power conditions.

Understanding the behavior and limitations of batteries and capacitors made me appreciate why relying on them alone isn't always practical. Instead, combining smarter energy storage with strategies designed for unstable power makes it possible to build systems that are more efficient and reliable.

References

[1] Zhuangwei Shi. Mambalithium: Selective state space model for remaining-useful-life, state-of-health, and state-of-charge estimation of lithium-ion batteries. *arXiv preprint arXiv:2403.05430*, 2024.